

ON COMPARING TWO SIMPLE LINEAR REGRESSION LINES

F.E. Steffens

National Research Institute for Mathematical Sciences, Pretoria

S U M M A R Y

Various tests are discussed by means of which two simple linear regression lines may be tested for differences in slope and position, using the Bonferroni inequality and Scheffé's F-projections. The latter method allows the real line to be divided into two subregions such that the null hypothesis of equal lines is rejected in the one subregion (which may be empty) and not in the other. These methods are extended to the case of unequal variances using Welch's APDF solutions. The various methods are compared by means of a Monte Carlo study.

1. Introduction

Comparing two lines by the usual analysis of covariance method in which the lines are assumed to be parallel is not always satisfactory, mainly because the assumption of parallelism may not always be feasible. It may often be more convenient to say that the two lines are significantly different in a certain region, while outside the region the two lines may possibly not be different. The region of non-rejection may be too far away from the observation points or it may contain a point of intersection with a certain probability. Some of the tests given here are not new, but they are referred to in the simulation study, which makes it convenient to include them in the text.

The discussions may be extended to multivariate regression functions and/or more than two functions. This is not done in the present paper because many interesting and useful results are obscured by notational and algebraic difficulties in the general case. An extension to polynomials is intended.

The expressions $t(f)$ and $F(f_1, f_2)$ are used to denote Student's t-random variable with f degrees of freedom and the F-random variable of Snedecor with f_1 and f_2 degrees of freedom respectively. The signs " $\sim$ " and " $\approx$ " are read "is distributed as" and "is approximately distributed as". By $t(f, \alpha)$ will be denoted the upper two-sided α -critical value of Student's t with f degrees of freedom, and by $F(f_1, f_2, \alpha)$ the upper α -critical value of F with f_1 and f_2 degrees of freedom.

2. A simultaneous test for position and direction

2.1 Given are two sets of data

$$(x_{ij}, y_{ij}), j=1, \dots, n_i; i=1, 2,$$

where the x_{ij} are chosen values of the non-random variable x , and the y_{ij} are observed values of the random variable y , the expected value of y_{ij} depending on i and x_{ij} .

Define

$$\bar{x}_i = \sum_j x_{ij}/n_i, e_i = \sum_j (x_{ij} - \bar{x}_i)^2.$$

$n_i, \bar{x}_i$ and e_i are determined by the plan of the experiment, i.e. by the position and number of points at which observations are taken.

2.2 Let

$$\underline{Y}' = (y_{11}, \dots, y_{1n_1}, y_{21}, \dots, y_{2n_2}),$$

$$\underline{X}' = \begin{bmatrix} 1 & \dots & 1 & 0 & \dots & 0 \\ x_{11} & \dots & x_{1n_1} & 0 & \dots & 0 \\ 0 & \dots & 0 & 1 & \dots & 1 \\ 0 & \dots & 0 & x_{21} & \dots & x_{2n_2} \end{bmatrix}$$

$$\underline{\beta}' = (\beta_{10}, \beta_{11}, \beta_{20}, \beta_{21}).$$

It is assumed that

$$(2.2.1) \quad \underline{Y} \sim N(\underline{X}\underline{\beta}, \sigma^2 \underline{I}).$$

The least squares estimate $\hat{\underline{\beta}}$ of $\underline{\beta}$ is given by

$$\underline{b}' = \hat{\underline{\beta}}' = (b_{10}, b_{11}, b_{20}, b_{21})$$

where

$$(2.2.2) \quad b_{i1} = \sum_j y_{ij}(x_{ij} - \bar{x}_i)/e_i, b_{i0} = \bar{y}_i - b_{i1}\bar{x}_i, \\ \bar{y}_i = \sum_j y_{ij}/n_i,$$

or

$$(2.2.3) \quad \underline{b} = (\underline{X}'\underline{X})^{-1}\underline{X}'\underline{Y}.$$

The unbiased estimate of σ^2 is

$$s^2 = \hat{\sigma}^2 = \sum_{ij} (y_{ij} - b_{i0} - b_{i1}x_{ij})^2 / (n_1 + n_2 - 4).$$

From (2.2.1) and (2.2.3) it follows that $\underline{b} \sim N(\underline{\beta}, \sigma^2(\underline{X}'\underline{X})^{-1})$ with $\underline{X}'\underline{X} = \begin{bmatrix} \underline{A}_1 & \underline{0} \\ \underline{0} & \underline{A}_2 \end{bmatrix}$ where $\underline{A}_i = \begin{bmatrix} n_i & \sum_j x_{ij} \\ \sum_j x_{ij} & \sum_j x_{ij}^2 \end{bmatrix}$

Therefore

$$(2.2.4) \quad (\underline{X}'\underline{X})^{-1} = \begin{bmatrix} \underline{v}_1 & \underline{0} \\ \underline{0} & \underline{v}_2 \end{bmatrix}$$

where $\underline{v}_i = \underline{A}_i^{-1} =$

$$(2.2.5) \quad \begin{bmatrix} 1/n_i + \bar{x}_i^2/e_i & -\bar{x}_i/e_i \\ -\bar{x}_i/e_i & 1/e_i \end{bmatrix}$$

It follows that

$$\begin{aligned}
 \text{Var } (b_{10}-b_{20}) &= \sigma^2 (1/n_1 + \bar{x}_1^2/e_1 + 1/n_2 + \bar{x}_2^2/e_2) \\
 (2.2.6) \quad \text{Var } (b_{11}-b_{21}) &= \sigma^2 (1/e_1 + 1/e_2) \\
 \text{Cov } (b_{10}-b_{20}, b_{11}-b_{21}) &= \sigma^2 (-\bar{x}_1/e_1 - \bar{x}_2/e_2).
 \end{aligned}$$

The sample estimates $\hat{\text{Var}} (b_{10}-b_{20})$, $\hat{\text{Var}} (b_{11}-b_{21})$, $\hat{\text{Cov}} (b_{10}-b_{20}, b_{11}-b_{21})$ are obtained by replacing σ^2 by s^2 in (2.2.6).

2.3 Define $X_o = (e_2 \bar{x}_1 + e_1 \bar{x}_2)/(e_1 + e_2)$, i.e. X_o is a point between $\bar{x}_1$ and $\bar{x}_2$, being nearer to the $\bar{x}_i$ corresponding to the smaller e_i . X_o lies midway between $\bar{x}_1$ and $\bar{x}_2$ if $e_1 = e_2$, and $X_o = \bar{x}$ if $\bar{x} = \bar{x}_1 = \bar{x}_2$.

By means of (2.2.6) X_o can also be written as

$$(2.3.1) \quad X_o = -\text{Cov } (b_{10}-b_{20}, b_{11}-b_{21}) / \text{Var } (b_{11}-b_{21}).$$

Theorem 1. X_o is the point such that

(2.3.2) $\text{Var } \{(b_{10}+b_{11}x)-(b_{20}+b_{21}x)\}$ has its minimum in X_o , and

(2.3.3) $\text{Cov } (b_{10}+b_{11}x-b_{20}-b_{21}x, b_{11}-b_{21}) = 0$ if $x=X_o$.

Proof. Let

$$\begin{aligned}
 v &= \text{Var } (b_{10}+b_{11}x-b_{20}-b_{21}x) \\
 &= \text{Var } (b_{10}-b_{20}) + 2x \text{Cov } (b_{10}-b_{20}, b_{11}-b_{21}) + x^2 \text{Var } (b_{11}-b_{21}).
 \end{aligned}$$

From (2.3.1), $\partial v / \partial x = 0$ if $x=X_o$. Hence (2.3.2) follows.

Also,

$$\begin{aligned}
 \text{Cov } (b_{10}+b_{11}x-b_{20}-b_{21}x, b_{11}-b_{21}) &= \text{Cov } (b_{10}-b_{20}, b_{11}-b_{21}) \\
 &\quad + x \text{Var } (b_{11}-b_{21}) \\
 &= 0 \text{ if } x=X_o, \text{ from (2.3.1).}
 \end{aligned}$$

2.4 Define

$$(2.4.1) \quad r = \frac{\text{Cov } (b_{10}-b_{20}, b_{11}-b_{21})}{\sqrt{\text{Var } (b_{10}-b_{20}) \text{Var } (b_{11}-b_{21})}}$$

and

$$(2.4.2) \quad \frac{d}{d\sigma} = \left[\frac{b_{10}-b_{20}}{\sqrt{\text{Var } (b_{10}-b_{20})}} - r \frac{b_{11}-b_{21}}{\sqrt{\text{Var } (b_{11}-b_{21})}} \right] / \sqrt{1-r^2}.$$

Using (2.3.1) and (2.4.1) the last expression can be written

$$(2.4.3) \quad \frac{d}{d\sigma} = \{b_{10}-b_{20} + (b_{11}-b_{21}) X_o\} / \sqrt{\text{Var } (b_{10}-b_{20})(1-r^2)}.$$

Theorem 2.

$$(2.4.4) \quad \text{Cov} (d, b_{11}-b_{21}) = 0, \text{ and}$$

$$(2.4.5) \quad \text{Var} (d) = \sigma^2.$$

Proof. (2.4.4) follows directly from (2.3.3) and (2.4.3). Using (2.4.2) and (2.4.1)

$$\text{Var} \left(\frac{d}{\sigma} \right) = (1-2r^2+r^2)/(1-r^2) = 1,$$

so that (2.4.5) follows.

Remark

$$(2.4.6) \quad d \sim N(\text{Ed}, \sigma^2)$$

where

$$E\left(\frac{d}{\sigma}\right) = \{\beta_{10}-\beta_{20} + (\beta_{11}-\beta_{21}) X_0\} / \sqrt{\text{Var} (b_{10}-b_{20})(1-r^2)}.$$

Theorem 3. Under the null hypothesis

$$H_0(1) : \beta_{10} + \beta_{11} X_0 = \beta_{20} + \beta_{21} X_0; \beta_{11} = \beta_{21}$$

the variable

$$\frac{1}{2} \left[\frac{d^2}{s^2} + \frac{(b_{11}-b_{21})^2}{s^2 \left(\frac{1}{e_1} + \frac{1}{e_2} \right)} \right] \sim F(2, n_1+n_2-4).$$

Proof. This follows from the fact that d and $(b_{11}-b_{21})/(1/e_1+1/e_2)^{1/2}$ are independently distributed as $N(0, \sigma^2)$ under $H_0(1)$, using (2.4.6), (2.4.4) and (2.2.6).

Remark. Thus $H_0(1)$ can be rejected at the α -level of significance if

$$(2.4.7) \quad \frac{d^2}{s^2} + \frac{(b_{11}-b_{21})^2}{s^2(1/e_1+1/e_2)} \geq 2F(2, n_1+n_2-4, \alpha).$$

3. Defining a rejection region

3.1 The following theorem on Scheffé F-projections is proved *inter alia* in Miller (1966) p. 65:

Theorem 4. If $Y \sim N(X\beta, \sigma^2 I)$ where X is an $n \times p$ matrix of rank $p < n$ and L is a d_1 -dimensional subspace of the p -dimensional Euclidean space ($d_1 \leq p$) then

$$(3.1.1) \quad P\{|\underline{l}'(\underline{\hat{\beta}} - \underline{\beta})| < (d_1 F(d_1, n-p, \alpha))^{1/2} s (\underline{l}'(X'X)^{-1}\underline{l})^{1/2}, \forall \underline{l} \in L\} = 1-\alpha,$$

where s^2 is an unbiased estimate of σ^2 .

3.2 In the present case, choose $L = \{(u, x, -u, -x)', \forall u, x\}$.

An orthonormal basis for L is $(1/\sqrt{2}, 0, -1/\sqrt{2}, 0)', (0, 1/\sqrt{2}, 0, -1/\sqrt{2})'$, i.e. $d_1=2$.

$n=n_1+n_2, p=4$ if each line has been observed at at least two different values of x , and $n>4$. By choosing different combinations

of u and x a series of hypotheses of the general form

$$H_0(2) : u\beta_{10} + x\beta_{11} = u\beta_{20} + x\beta_{21}$$

may be tested simultaneously.

Using (2.2.4) and (2.2.5) it is seen that, with $\underline{u} = (u, x, -u, -x)'$,

$$\underline{u}'(\underline{X}'\underline{X})^{-1}\underline{u} = u^2/n_1 + (u\bar{x}_1 - x)^2/e_1 + u^2/n_2 + (u\bar{x}_2 - x)^2/e_2.$$

Thus from (3.1.1) with $u=0$, $x=1$, the null hypothesis

$$H_0(3) : \beta_{11} = \beta_{21}$$

is rejected at the α -level if

$$(3.2.1) \quad (b_{11} - b_{21})^2 \geq 2s^2 F(2, n_1 + n_2 - 4, \alpha)(1/e_1 + 1/e_2).$$

Choosing $u=1$, $x=0$, the null hypothesis

$$H_0(4) : \beta_{10} = \beta_{20}$$

is rejected at the α -level if

$$(3.2.2) \quad (b_{10} - b_{20})^2 \geq 2s^2 F(2, n_1 + n_2 - 4, \alpha)(1/n_1 + \bar{x}_1^2/e_1 + 1/n_2 + \bar{x}_2^2/e_2).$$

Choosing $u=1$, $x=x$, the null hypothesis

$$H_0(5) : \beta_{10} + \beta_{11}x = \beta_{20} + \beta_{21}x$$

is rejected at the α -level for all x such that

$$(3.2.3) \quad \{(b_{10} - b_{20}) + (b_{11} - b_{21})x\}^2 \geq 2s^2 F(2, n_1 + n_2 - 4, \alpha) \{1/n_1 + (x - \bar{x}_1)^2/e_1 + 1/n_2 + (x - \bar{x}_2)^2/e_2\}.$$

3.3 Writing

$$g^2 = 2F(2, n_1 + n_2 - 4, \alpha)$$

the inequality (3.2.3) may be rewritten as

$$(3.3.1) \quad \{(b_{11} - b_{21})^2 - g^2 \text{Var}(b_{11} - b_{21})\}x^2 + 2\{(b_{11} - b_{21})(b_{10} - b_{20}) - g^2 \text{Cov}(b_{11} - b_{21}, b_{10} - b_{20})\}x + \{(b_{10} - b_{20})^2 - g^2 \text{Var}(b_{10} - b_{20})\} \geq 0.$$

Solving for x , two values X_1 and X_2 are found such that, if they are real, $H_0(5)$ is rejected either inside or outside (X_1, X_2) .

If (3.2.1) holds, i.e. the hypothesis of parallelism is rejected, the coefficient of x^2 in (3.3.1) is positive and $H_0(5)$ is rejected outside (X_1, X_2) . If the hypothesis of parallelism is not rejected, $H_0(5)$ is rejected inside (X_1, X_2) . The discriminant D

of the inequality (3.3.1) is given by

$$D^2/4 = g^2 \text{Var}(b_{10} - b_{20}) \text{Var}(b_{11} - b_{21}) \left\{ \left(\frac{b_{10} - b_{20}}{\sqrt{\text{Var}(b_{10} - b_{20})}} - r \frac{b_{11} - b_{21}}{\sqrt{\text{Var}(b_{11} - b_{21})}} \right)^2 + \frac{(b_{11} - b_{21})^2}{\text{Var}(b_{11} - b_{21})} (1 - r^2) - g^2 (1 - r^2) \right\}, \text{ using (2.4.1),}$$

$$(3.3.2) \quad = g^2(1-r^2)\text{V}\bar{\text{a}}\text{r}(b_{10}-b_{20})\text{V}\bar{\text{a}}\text{r}(b_{11}-b_{21})\left\{\frac{d^2}{s^2} + \frac{(b_{11}-b_{21})^2}{\text{V}\bar{\text{a}}\text{r}(b_{11}-b_{21})} - g^2\right\}$$

using (2.4.2).

If $D^2 \geq 0$, the two values (X_1, X_2) will be real. Note that $D^2 \geq 0$ if and only if (2.4.7) is satisfied, i.e. if $H_0(1)$ is rejected by means of the simultaneous test in §2. Inspection of (3.3.2) shows that, if $H_0(3)$ is rejected, i.e. if the lines are not parallel, $D^2 \geq 0$ and X_1, X_2 are always real. If $H_0(3)$ is not rejected, X_1 and X_2 may be complex numbers.

The collection of values of x for which $H_0(5)$ is rejected is called the rejection region. The rejection region may be empty, the interval (X_1, X_2) or its complement. The interpretation of the region of non-rejection may either be that it constitutes an interval within which the population lines intersect with probability $1-\alpha$ (see e.g. Kastenbaum (1959)), or that the values of x in the region are too far away from X_0 for a difference between the regression values to be shown out by means of the available data. If the rejection region is empty, the population lines may be identical or more data is needed to show a difference to exist. The point of intersection of the estimated regression lines is always in the region of non-rejection (left hand side of (3.2.3) equal to 0).

3.4 The logical procedure would seem to be as follows:

- (a) Perform test (2.4.7). If $H_0(1)$ is not rejected, the tests in §3 will all be negative.
- (b) If $H_0(1)$ is rejected, $H_0(3)$ and $H_0(4)$ may be tested to see whether the lines differ in slope or position ((3.2.1) and (3.2.2)).
- (c) The inequality (3.3.1) will have real roots if $H_0(1)$ is rejected, the interpretation of (X_1, X_2) being dictated by the rejection or not of $H_0(3)$.

A resolution into a rejection region and a non-rejection region has been suggested in Walker and Lev (1953) p. 399, but there the critical value is $t(n_1+n_2-4, \alpha)$ instead of $2F(2, n_1+n_2-4, \alpha)$.

4. Other tests on the differences between the coefficients

4.1 If the investigator is not interested in defining a rejection region but merely in comparing the regression coefficients, he may do better by using the Bonferroni inequality (see e.g. Miller (1966) p. 8) rather than Scheffé F-projections.

Thus $H_0(3) : \beta_{11} = \beta_{21}$ is rejected at the α -level if

$$(4.1.1) \quad |b_{11} - b_{21}| \geq st(n_1+n_2-4, \frac{1}{2}\alpha)(1/e_1 + 1/e_2)^{\frac{1}{2}}$$

and $H_0(4) : \beta_{10} = \beta_{20}$ is rejected at the α -level if

$$(4.1.2) \quad |b_{10} - b_{20}| \geq t(n_1 + n_2 - 4, \frac{1}{2}\alpha) \{1/n_1 + \bar{x}_1^2/e_1 + 1/n_2 + \bar{x}_2^2/e_2\}^{\frac{1}{2}}.$$

If only one of the two tests is performed, the critical constant in (4.1.1) or (4.1.2) may be lowered to $t(n_1 + n_2 - 4, \alpha)$.

4.2 If the investigator is sure or feels justified in assuming that

$$(4.2.1) \quad \beta_{11} = \beta_{21} = \beta_1$$

he may use that information by fitting parallel lines directly (see e.g. Brownlee (1960) p. 288):

$$\hat{b}_1 = \hat{\beta}_1 = \sum_{ij} y_{ij} (x_{ij} - \bar{x}_1) / (e_1 + e_2),$$

$$\hat{b}_{i0} = \hat{\beta}_{i1} = \bar{y}_i - \hat{b}_1 \bar{x}_i,$$

$$s_p^2 = \hat{\sigma}^2 = \sum_{ij} (y_{ij} - \hat{b}_{i0} - \hat{b}_1 x_{ij})^2 / (n_1 + n_2 - 3), \text{ and}$$

$$\text{Var}(\hat{b}_{10} - \hat{b}_{20}) = \sigma^2 \{1/n_1 + 1/n_2 + (\bar{x}_1 - \bar{x}_2)^2 / (e_1 + e_2)\}.$$

Then $H_0(4) : \beta_{10} = \beta_{20}$ is rejected at the α -level if

$$|\hat{b}_{10} - \hat{b}_{20}| \geq s_p t(n_1 + n_2 - 3, \alpha) \{1/n_1 + 1/n_2 + (\bar{x}_1 - \bar{x}_2)^2 / (e_1 + e_2)\}^{\frac{1}{2}}$$

However, if this test is preceded by a test whether (4.2.1) is valid (as in (4.1.1)), then both tests must be performed at the $\frac{1}{2}\alpha$ -level to ensure that the simultaneous significance level is less than or equal to α .

5. The case of unequal variances

5.1 The assumptions of section 2.2 remain the same, except that (2.2.1) is replaced by

$$\underline{Y} \sim N(\underline{X}\underline{\beta}, \underline{\Sigma})$$

where

$$\underline{\Sigma} = \begin{bmatrix} \sigma_1^2 & & & & \\ & \ddots & & & \\ & & \sigma_1^2 & & 0 \\ & 0 & & \sigma_2^2 & \\ & & & & \ddots \\ & & & & & \sigma_2^2 \end{bmatrix}$$

It is sometimes assumed that

$$(5.1.1) \quad \sigma_1^2 = \sigma_2^2 = \sigma^2/m$$

with m known, i.e. $\underline{\Sigma} = \sigma^2 \underline{T}$,

$$\text{with } \underline{T} = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & 0 \\ & 0 & & \ddots & \\ & & & & m \end{bmatrix}$$

The sample estimate $\underline{b}$ of $\underline{\beta}$ remains the same as in (2.2.2), but now

$$\text{Var } \underline{b} = \begin{bmatrix} \sigma_1^2 \underline{v}_1 & 0 \\ 0 & \sigma_2^2 \underline{v}_2 \end{bmatrix}$$

where $\underline{v}_i$ is given by (2.2.5).

Thus

$$\text{Var}(b_{j0} - b_{20}) = \sigma_1^2 (1/n_1 + \bar{x}_1^2/e_1) + \sigma_2^2 (1/n_2 + \bar{x}_2^2/e_2),$$

$$(5.1.2) \quad \text{Var}(b_{11} - b_{21}) = \sigma_1^2 (1/e_1) + \sigma_2^2 (1/e_2),$$

$$\text{Cov}(b_{10} - b_{20}, b_{11} - b_{21}) = \sigma_1^2 (-\bar{x}_1/e_1) + \sigma_2^2 (-\bar{x}_2/e_2).$$

The two major results used here are summarized as theorems 5 and 6.

Theorem 5. (Miller (1966) p. 53): If $\underline{Y} \sim N(\underline{X}\underline{B}, \sigma^2 \underline{T})$, $\underline{T}$ known and nonsingular, $\underline{X}$ is an $n \times p$ matrix of rank p ($p < n$) and $\underline{L}$ is a d_1 -dimensional subspace of the p -dimensional Euclidean space ($d_1 \leq p$), then

$$P\{\underline{L}'\underline{B}\epsilon\epsilon'\underline{B} \pm \{d_1 F(d_1, n-p, \alpha)\}^{\frac{1}{2}} s\{\underline{L}'(\underline{X}'\underline{T}^{-1}\underline{X})^{-1}\underline{L}\}^{\frac{1}{2}}, \forall \underline{L} \in \underline{L}\} \\ = 1 - \alpha,$$

where s^2 is an unbiased estimate of σ^2 .

Theorem 6. (Welch (1947)): Let η be a population parameter which is estimated by an observed quantity y , normally distributed with variance $\sigma_y^2 = \sum \lambda_i \sigma_i^2$ where the λ_i are known positive numbers and the σ_i^2 are unknown variances. Suppose that the observed data provide estimates s_i^2 of the variances σ_i^2 based on f_i' degrees of freedom respectively. Then

$$(y - \eta) / (\sum \lambda_i s_i^2)^{\frac{1}{2}}$$

is approximately distributed as Student's t with degrees of freedom

$$(5.1.3) \quad f = (\sum \lambda_i \sigma_i^2)^2 / \sum (\lambda_i^2 \sigma_i^4 / f_i').$$

If nothing is known about the relative sizes of the σ_i^2 , f may be estimated by

$$(5.1.4) \quad \hat{f} = (\sum \lambda_i s_i^2)^2 / \sum \{\lambda_i^2 s_i^4 / (f_i' + 2)\} - 2$$

5.2 In order to apply theorem 5 it is assumed that m in (5.1.1) is known. The sample estimate of σ^2 is

$$s_u^2 = \sigma^2 = \{\Sigma (y_{1j} - b_{10} - b_{11}x_{1j})^2 + \Sigma (y_{2j} - b_{20} - b_{21}x_{2j})^2 / m\} / (n_1 + n_2 - 4).$$

Also, (5.1.2) becomes

$$\text{Var}(b_{10} - b_{20}) = \sigma^2 (1/n_1 + \bar{x}_1^2/e_1 + m/n_2 + m\bar{x}_2^2/e_2),$$

$$(5.2.1) \quad \text{Var}(b_{11} - b_{21}) = \sigma^2 (1/e_1 + m/e_2),$$

$$\text{Cov}(b_{10}-b_{20}, b_{11}-b_{21}) = \sigma^2(-\bar{x}_1/e_1 - m\bar{x}_2/e_2).$$

The sample estimates of these quantities are found by replacing σ^2 in (5.2.1) by s_u^2 . Bearing these alterations in mind, the procedure of section 3.4 is adaptable to the present situation without further difficulties. Thus e.g. $H_0(5) : \beta_{10} + \beta_{11}x = \beta_{20} + \beta_{21}x$ is rejected at the α -level for all x such that

$$(5.2.2) \quad \{(b_{10}-b_{20}) + (b_{11}-b_{21})x\}^2 \geq 2s_u^2 F(2, n_1+n_2-4, \alpha).$$

$$\cdot \{1/n_1 + (x-\bar{x}_1)^2/e_1 + m/n_2 + m(x-\bar{x}_2)^2/e_2\}.$$

From this inequality X_1, X_2 may again be found such that $H_0(5)$ is rejected either inside or outside (X_1, X_2) .

The case m unknown will be treated slightly differently and under stricter conditions in section 5.4.

Separate tests on the slopes and abscissae as in sec. 4.1 may also be based on (5.2.1).

5.3 If separate tests are to be performed simultaneously on $\beta_{10}-\beta_{20}$ and $\beta_{11}-\beta_{21}$, theorem 6 may be applied.

Firstly, using (5.1.3) with $\lambda_i = 1/e_i$ and (5.1.2),

$$\{(b_{11}-b_{21}) - (\beta_{11}-\beta_{21})\} / (s_1^2/e_1 + s_2^2/e_2)^{1/2} \sim t(f_1)$$

where

$$s_i^2 = \sum_j (y_{ij} - b_{i0} - b_{i1}x_{ij})^2 / (n_i - 2)$$

and

$$(5.3.1) \quad f_1 = (n_1-2)(n_2-2)(m^2e_1^2 + e_2^2) / \{m^2e_1^2(n_1-2) + e_2^2(n_2-2)\}.$$

Therefore, reject $H_0(3) : \beta_{11} = \beta_{21}$ at the α -level if

$$(5.3.2) \quad |b_{11}-b_{21}| \geq (s_1^2/e_1 + s_2^2/e_2)^{1/2} t(f_1, \frac{1}{2}\alpha).$$

If m is unknown, the application of (5.1.4) yields

$$(5.3.3) \quad \hat{f}_1 = n_1n_2(e_2s_1^2 + e_1s_2^2)^2 / (n_2e_2^2s_1^4 + n_1e_1^2s_2^4) - 2,$$

and f_1 in (5.3.2) must be replaced by $\hat{f}_1$ when testing $H_0(3)$.

Secondly, using (5.1.3) with $\lambda_i = 1/n_i + \bar{x}_i^2/e_i$ and (5.1.1),

$$(5.3.4) \quad \{(b_{10}-b_{20}) - (\beta_{10}-\beta_{20})\} / \{s_1^2(1/n_1 + \bar{x}_1^2/e_1) + s_2^2(1/n_2 + \bar{x}_2^2/e_2)\}^{1/2} \sim t(f_2)$$

with

$$(5.3.5) \quad f_2 = \frac{\{(1/n_1 + \bar{x}_1^2/e_1) + m(1/n_2 + \bar{x}_2^2/e_2)\}^2}{(1/n_1 + \bar{x}_1^2/e_1)^2/(n_1-2) + m^2(1/n_2 + \bar{x}_2^2/e_2)^2/(n_2-2)}.$$

Therefore, reject $H_0(4) : \beta_{10} = \beta_{20}$ at the α -level if

$$(5.3.6) \quad |b_{10} - b_{20}| \geq \{s_1^2(1/n_1 + \bar{x}_1^2/e_1) + s_2^2(1/n_2 + \bar{x}_2^2/e_2)\}^{1/2} t(f_2, 1/\alpha).$$

Again, if m is unknown, f_2 is estimated, using (5.1.4), by

$$(5.3.7) \quad \hat{f}_2 = \frac{\{s_1^2(1/n_1 + \bar{x}_1^2/e_1) + s_2^2(1/n_2 + \bar{x}_2^2/e_2)\}^2}{s_1^4(1/n_1 + \bar{x}_1^2/e_1)^2/n_1 + s_2^4(1/n_2 + \bar{x}_2^2/e_2)^2/n_2} - 2,$$

and f_2 is replaced by $\hat{f}_2$ in (5.3.6) when $H_0(4)$ is tested.

If only one of $H_0(3)$ or $H_0(4)$ is tested, the critical constant in (5.3.2) or (5.3.6) may be lowered to $t(f_1, \alpha)$ or $t(f_2, \alpha)$ respectively.

5.4 A variation in the proof of theorem 5 as given in Miller (1966) p. 53, makes the method of section 5.2 available in the case m unknown, but under certain conditions on the plan of the experiment. These conditions are as follows:

$$(5.4.1) \quad \bar{x} = \bar{x}_1 = \bar{x}_2, \quad n = n_1 = kn_2, \quad e = e_1 = ke_2$$

where k is a positive real number. Such a case would arise e.g. if the observation points are the same for the two lines, with k_1 observations per point on the first line and k_2 observations per point on the second line, and $k = k_1/k_2$.

Let

$$\underline{L} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix},$$

$$\underline{Y} = \underline{L} \underline{\beta} = (\beta_{10} - \beta_{20}, \beta_{11} - \beta_{21})', \text{ and}$$

$$\hat{\underline{Y}} = \underline{L} \underline{b} = (b_{10} - b_{20}, b_{11} - b_{21})'.$$

$$\text{Then } \underline{\hat{Y}} \sim N(\underline{Y}, \underline{L} \text{ Var } \underline{b} \underline{L}')$$

$$\text{where } \underline{L} \text{ Var } \underline{b} \underline{L}' = \sigma_1^2 \underline{V}_1 + \sigma_2^2 \underline{V}_2$$

$$= \begin{bmatrix} \text{Var}(b_{10} - b_{20}) & \text{Cov}(b_{10} - b_{20}, b_{11} - b_{21}) \\ \text{Cov}(b_{10} - b_{20}, b_{11} - b_{21}) & \text{Var}(b_{11} - b_{21}) \end{bmatrix}$$

In the special case (5.4.1) this can be written, using (5.1.2),

$$\underline{L} \text{ Var } \underline{b} \underline{L}' = (\sigma_1^2 + k\sigma_2^2) \begin{bmatrix} 1/n + \bar{x}^2/e & -\bar{x}/e \\ -\bar{x}/e & 1/e \end{bmatrix}.$$

Now, let

$$\underline{P} = \begin{bmatrix} \sqrt{n} & \sqrt{n\bar{x}} \\ 0 & \sqrt{e} \end{bmatrix}.$$

Then

$$\underline{\hat{\delta}} = \underline{P} \underline{\hat{Y}} = (\sqrt{n}(b_{10} - b_{20}) + \sqrt{n\bar{x}}(b_{11} - b_{21}), \sqrt{e}(b_{11} - b_{21}))'$$

$$\sim N(\underline{P} \underline{Y}, \underline{P} \underline{L} \text{ Var } \underline{b} \underline{L}' \underline{P}')$$

$$\text{where } \underline{P} \underline{L} \text{ Var } \underline{b} \underline{L}' \underline{P}' = (\sigma_1^2 + k\sigma_2^2) \underline{I}.$$

Applying Welch's APDF solution (theorem 6), (assuming m known), it is seen that, under

$$H_0(6) : (\beta_{10} - \beta_{20}) + \bar{x}(\beta_{11} - \beta_{21}) = 0; \beta_{11} - \beta_{21} = 0,$$

(which is exactly $H_0(1)$ in this special case),

$$\{\sqrt{n}(b_{10} - b_{20}) + \sqrt{nx}(b_{11} - b_{21})\} / (s_1^2 + ks_2^2)^{\frac{1}{2}}$$

$$\text{and } \sqrt{e}(b_{11} - b_{21}) / (s_1^2 + ks_2^2)^{\frac{1}{2}}$$

are each approximately distributed as $t(f_3)$, where

$$(5.4.2) \quad f_3 = (1 + km)^2 / \{1 / (n-2) + k^3 m^2 / (n-2k)\}.$$

The numerators are independent, so that

$$(5.4.3) \quad \frac{1}{2} [n\{(b_{10} - b_{20}) + \bar{x}(b_{11} - b_{21})\}^2 + e(b_{11} - b_{21})^2] / (s_1^2 + ks_2^2) \\ \sim F(2, f_3).$$

Therefore $H_0(6)$ is rejected at the α -level if

$$n\{(b_{10} - b_{20}) + \bar{x}(b_{11} - b_{21})\}^2 + e(b_{11} - b_{21})^2 \geq 2(s_1^2 + ks_2^2) F(2, f_3, \alpha).$$

5.5 By an argument similar to that in Miller (1966) p. 65 it can be shown that

$$n\{(b_{10} - b_{20}) - (\beta_{10} - \beta_{20}) + \bar{x}(b_{11} - b_{21}) - \bar{x}(\beta_{11} - \beta_{21})\}^2 \\ + e\{(b_{11} - b_{21}) - (\beta_{11} - \beta_{21})\}^2 < 2(s_1^2 + ks_2^2) F(2, f_3, \alpha)$$

is true if and only if

$$[u\{(b_{10} - b_{20}) - (\beta_{10} - \beta_{20})\} + x\{(b_{11} - b_{21}) - (\beta_{11} - \beta_{21})\}]^2 \\ < 2(s_1^2 + ks_2^2) \{u^2 / n + (u\bar{x} - x)^2 / e\} F(2, f_3, \alpha), \forall u, x.$$

Thus $H_0(3) : \beta_{11} \neq \beta_{21}$ is rejected at the α -level if

$$e(b_{11} - b_{21})^2 \geq 2(s_1^2 + ks_2^2) F(2, f_3, \alpha),$$

$H_0(4) : \beta_{10} - \beta_{20}$ is rejected at the α -level if

$$(b_{10} - b_{20})^2 \geq 2(1/n + \bar{x}^2/e)(s_1^2 + ks_2^2) F(2, f_3, \alpha),$$

and $H_0(5) : \beta_{10} + \beta_{11}x = \beta_{20} + \beta_{21}x$ is rejected at the α -level for x such that

$$(5.5.1) \quad \{(b_{10} - b_{20}) + (b_{11} - b_{21})x\}^2 \geq 2(s_1^2 - ks_2^2) \{1/n + (x - \bar{x})^2/e\} F(2, f_3, \alpha)$$

Thus again two values X_1, X_2 may be found such that $H_0(5)$ is rejected either inside or outside (X_1, X_2) .

If m is unknown, f_3 is estimated (using (5.1.4)) by

$$(5.5.2) \quad \hat{f}_3 = n(s_1^2 + ks_2^2)^2 / (s_1^4 + k^3 s_2^4) - 2.$$

In this case f_3 is replaced by $\hat{f}_3$ in the text from (5.4.3) to (5.5.1).

It is easily shown that, under conditions (5.4.1),

$$f_1 = f_2 = f_3, \quad \hat{f}_1 = \hat{f}_2 = \hat{f}_3$$

from (5.3.1), (5.3.3), (5.3.5), (5.3.7), (5.4.2) and (5.5.2).

5.6 Accuracy of the approximations

An important question which remains unanswered is the following: for which values of n_1, n_2 and k are the various approximate distributions accurate enough for application in practice? This question was pointed out by the editor, and will hopefully be answered in the not too distant future.

6. Simulation Results

6.1 In order to compare the tests described in this paper, a Monte Carlo study was performed on the IBM 360/40 computer of the C.S.I.R. Pseudo-random numbers $\{r_i\}$ uniformly distributed between 0 and 1 were obtained by the power residue method (see e.g. Jansson (1966) p. 76):

$$r_{i+1} \equiv r_i(2^{16}+3) \bmod (2^{31})$$

using the IBM-supplied routine RANDU.

Pseudo-random numbers $\{q_i\}$ with $N(0,1)$ -distribution were obtained from $\{r_i\}$ by means of the transformation of Box and Muller (1958):

$$q_i = (-2 \ln r_i)^{\frac{1}{2}} \cos 2\pi r_{i+1}$$

$$q_{i+1} = (-2 \ln r_i)^{\frac{1}{2}} \sin 2\pi r_{i+1}.$$

6.2 The two population lines to be compared were chosen (arbitrarily) as follows:

$$E(y_{1j}) = 1+2x, \quad \text{Var}(y_{1j}) = 1, \quad E(y_{2j}) = (1+\delta\beta_0)+(2+\delta\beta_1)x, \\ \text{Var}(y_{2j}) = m,$$

for various $\delta\beta_0, \delta\beta_1$ and m .

The observations were taken at $x=0, 0.5$ and 1 , with 2 observations per point on line 1 and 3 observations per point on line 2. Thus it is seen that (5.4.1) holds with $k=2/3$.

The proportion of times each null hypothesis is rejected when true is taken as an estimate of the true significance level of the test used to test it.

6.3 The tests performed on the two lines are summarized below for the sake of convenience.

A (Sec. 5.2): Separate tests for $\beta_{10}=\beta_{20}, \beta_{11}=\beta_{21}$ using a joint variance estimate $s_u^2 = \hat{\sigma}_1^2 = \hat{\sigma}_2^2/m$ (m known). Critical value $t(n_1+n_2-4, \alpha/2)$

B (Sec. 5.2): Joint test for $E(d)=0, \beta_{11}=\beta_{21}$ using s_u^2 as in A. Critical value $F(2, n_1+n_2-4, \alpha)$. If this test is

positive it may be followed by testing any combination $u(\beta_{10}-\beta_{20})+x(\beta_{11}-\beta_{21})=0$.

- C (Sec. 4.2): Usual analysis of covariance: a test for the slopes, followed by a test for the abscissae of parallel lines if the first test is negative. Variances assumed equal.
- D (Sec. 5.3): Tests for $\beta_{10}=\beta_{20}, \beta_{11}=\beta_{21}$ using Welch's APDF solution with m known. Critical values are $t(f_1, \alpha/2)$ and $t(f_2, \alpha/2)$.
- D' (Sec. 5.3): If m is unknown, f_1 and f_2 in D are replaced by $\hat{f}_1, \hat{f}_2$ respectively.
- E (Sec. 5.5): Test for $(\beta_{10}-\beta_{20})+\bar{x}(\beta_{11}-\beta_{21})=0$ using $F(2, f_3, \alpha)$. May be followed by tests for $u(\beta_{10}-\beta_{20})+x(\beta_{11}-\beta_{21})=0$. Applicable only if (5.4.1) holds and m is known.
- E' (Sec. 5.5): If m is unknown, f_3 in E is replaced by $\hat{f}_3$.

6.4 In order to find out what the effect of the choice of m is on the true significance levels of tests A, B, D and E, 1,000 samples each were generated for $m=16, m=1$ and $m=0.36$. The four tests were performed at the 10% level of significance for various guessed values of m , the null hypothesis that the lines are identical being true. The proportion of times that the null hypothesis was rejected when true may be interpreted as a sample estimate of the true significance level. The results are depicted in figure 1. From this graph two conclusions are evident:

- Tests A and B (based on joint estimates of σ^2) are influenced to a much larger extent by a wrong choice of m than tests D and E, which are based on Welch's APDF.
- Test A seems to have a smaller actual significance level than test B; similarly for tests D and E. Tests A and D are based on the Bonferroni inequality: to ensure that the actual significance level is at most 0.10, the tests on each set of coefficients is done at the 0.05 significance level.

6.5 To see which test is able to point out differences most easily, and what the influence of the choice of m is on that ability, 1,000 samples were generated with $m=4$ and various combinations of $\delta\beta_0 = \beta_{20}-\beta_{10}, \delta\beta_1 = \beta_{21}-\beta_{11}$. Some of these are depicted in figure 2, while the cases not depicted here follow exactly the same pattern. Once again it is seen that tests D and E are influenced to a smaller extent by the guessed value of m than A and B. Also, tests B and E showed real differences in a much larger proportion of the cases. At the true value of m there does not seem to be a great difference between A and D and between B and E.

6.6 The results of an investigation of the true level of significance of tests A, B, D, D', E and E' for various known values of m (i.e. assumed to be known exactly) are presented in table 1. The three tests based on the Bonferroni inequality seem to have true

significant levels in the vicinity of 0.08, while the true significance level of the APDF tests seem to be near 0.10, the nominal significance level. The two tests, D' and E', where m is assumed unknown, seem to have true significance levels which are higher than tests D and E respectively. In the case of test E', the true significance level seems to be definitely higher than 0.10. Investigation of the APDF's in sec. 5.3 and 5.5 shows that the maximum value of $\hat{f}_1, \hat{f}_2$ and $\hat{f}_3$ is n_1+n_2-2 , compared to n_1+n_2-4 for f_1, f_2 and f_3 . Thus the degrees of freedom seem to be overestimated, which may lead to a higher significance level than the nominal level.

6.7 In the case $m=1$, various sets of 1,000 samples were generated with different combinations of $\delta\beta_0 = \beta_{20} - \beta_{10}$, $\delta\beta_1 = \beta_{21} - \beta_{11}$. The results are presented in tables 2 and 3.

Table 2 gives the following results: For $\delta\beta_0 > 0$, $\delta\beta_1 = 0$ tests C seem to excel, especially for larger values of $\delta\beta_0$. This is of course the model for which test C is designed. Tests B and E seem to be not far behind, with B somewhat better than E. The performance of test E' is also good, but the possible overestimation of the degrees of freedom must be borne in mind. Where $\delta\beta_1 > 0$, test B seems to give the best performance, especially for larger values of $\delta\beta_1$.

Table 3 gives the number of times each test gave a significant outcome when testing $H_0(3)$ and $H_0(4)$. For $\delta\beta_0 > 0$, $\delta\beta_1 = 0$, test C gives the best performance by far, while A and D are next in line. In the case $\delta\beta_0 = 0$, $\delta\beta_1 > 0$ the test C shows a somewhat unexpected behaviour: In those cases where the tests could not show a difference between the slopes, it showed a difference between the abscissae (which did not exist) very easily. In this case test A seemed to be best. This is repeated in the case $\delta\beta_0 > 0$, $\delta\beta_1 > 0$, where test C is, strictly speaking, not applicable.

It is also seen that test D' (m unknown) is even somewhat better than D (m known) while it is nearly as good as test A. Thus it would seem best to use test D' when m is not known exactly and when the purpose is to find out whether the abscissae and/or the slopes are different.

7. Conclusion. From the Monte Carlo Study described in §6, it seems that the best test to use depends on the purpose and previous knowledge of the investigator:

- (a) if $m=1$, and the true lines are parallel the analysis of covariance test seems best.
- (b) if m is known but the lines are not known to be parallel, the rejection region technique with joint estimates of σ^2 seems best.
- (c) if m is known only approximately, the rejection region technique with APDF seems preferable. The test with m unknown needs some attention because the true significance level seems higher than the nominal level.

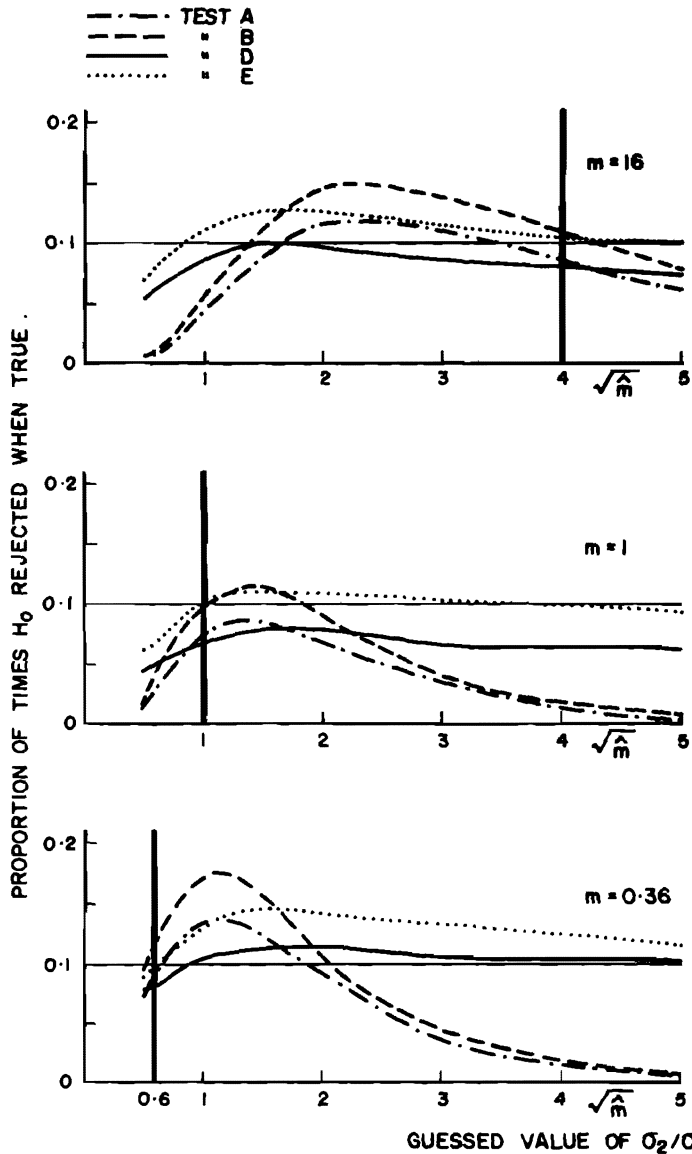


FIG.1 Effect of guessed value of m on the significance level of tests A,B,D and E for $m = 0.36, 1, 16$. Nominal significance level is 0.10, 1000 samples in each case.

Level 0.10 (10%)

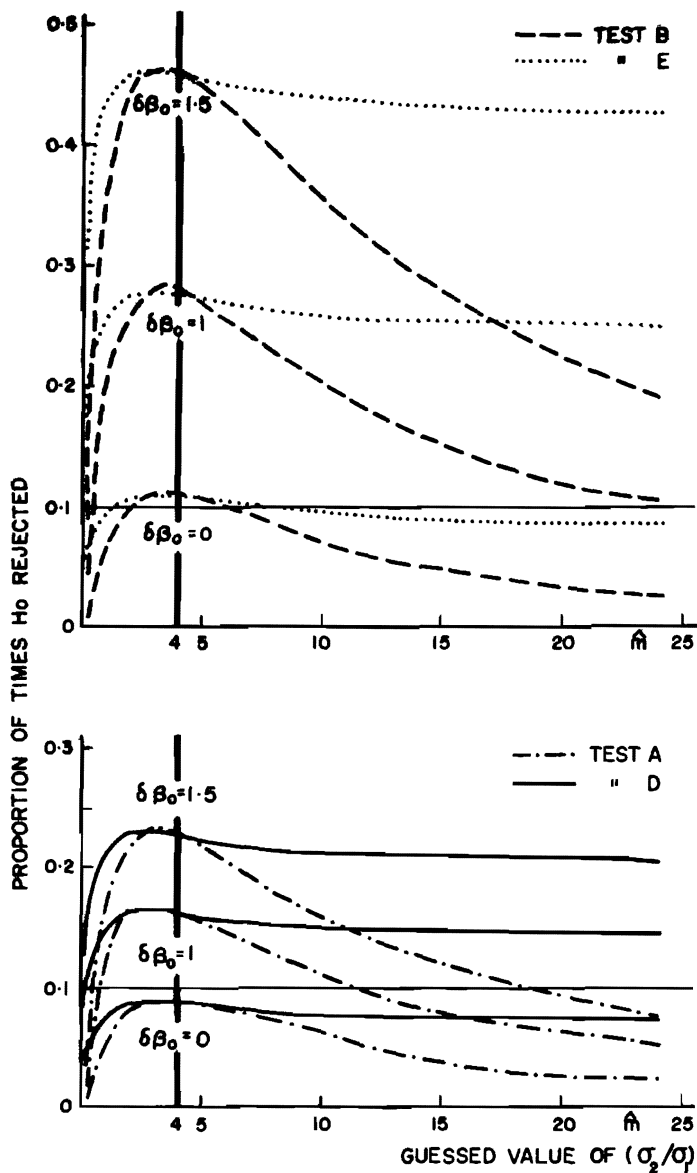


FIG. 2 Ability to reject H_0 by means of tests A, B, D, E when $m = 4$, $\beta_{21} = \beta_{11}$, $\delta\beta_0 = \beta_{20} - \beta_{10} = 0, 1, 1.5$. Significance level = 0.10. Each line is based on 1,000 samples, all tests being performed on the same samples for various guessed m .

TABLE 1

Number of times H_0 was rejected when true in 1,000 samples.
For each value of m a different set of samples was generated.
Significance level = 0.10.

Test m	A	B	D	E	D'	E'	C
25	85	97	77	96	83	102	82
16	84	110	81	104	90	118	
9	86	113	81	103	85	123	
4	79	112	77	111	82	124	
1	74	98	67	102	78	115	
0.81	68	96	71	100	82	116	
0.64	73	103	66	91	78	110	
0.49	80	94	80	90	90	112	
0.36	89	112	82	98	101	128	
0.25	107	121	92	122	113	147	
Total	825	1056	774	1017	882	1195	

TABLE 2

Number of times H_0 : the lines are identical was rejected by
means of different tests for various values of $\delta\beta_0 = \beta_{20} - \beta_{10}$,
 $\delta\beta_1 = \beta_{21} - \beta_{11}$ out of 1,000 samples. A different set of 1,000
samples was generated for each $(\delta\beta_0, \delta\beta_1)$. $\sigma_1^2 = \sigma_2^2 = 1$,
Significance level = 0.10.

		TEST						
$\delta\beta_0$	$\delta\beta_1$	A	B	C	D	D'	E	E'
0	0	74	107	109	71	77	106	123
0.5	0	109	184	182	105	119	184	224
1.0	0	206	429	428	205	220	425	430
1.5	0	406	745	762	386	400	721	728
2.0	0	612	914	930	589	608	904	911
0	0.5	86	146	139	89	99	151	179
0	1.0	138	233	214	133	143	226	244
0	1.5	205	410	367	198	208	396	422
0	2.0	320	619	567	294	302	596	614
0.5	0.5	138	325	317	135	145	321	347
1.0	1.0	268	763	753	252	267	728	749
1.5	1.5	554	991	977	521	535	971	973
2.0	2.0	776	999	998	747	758	998	998

TABLE 3

Number of times $H_0(4): \beta_{10} = \beta_{20}$ and $H_0(3): \beta_{11} = \beta_{21}$ was rejected out of 1,000 samples for various $\delta\beta_0 = \beta_{20} - \beta_{10}$, $\delta\beta_1 = \beta_{11} - \beta_{21}$ by means of different tests. The same samples as in table 2.

		TEST													
$\delta\beta_0$	$\delta\beta_1$	A		B		C		D		D'		E		E'	
		$H_O(4)$	$H_O(3)$	$H_O(4)$	$H_O(3)$	$H_O(4)$	$H_O(3)$	$H_O(4)$	$H_O(3)$	$H_O(4)$	$H_O(3)$	$H_O(4)$	$H_O(3)$	$H_O(4)$	$H_O(3)$
0	0	50	50	37	33	59	50	46	48	49	51	53	31	39	44
0.5	0	74	51	62	33	131	51	78	46	85	53	60	33	77	45
1.0	0	185	40	149	32	388	40	183	39	197	45	146	28	171	36
1.5	0	386	44	319	32	718	44	367	41	377	50	303	27	333	38
2.0	0	594	44	512	29	886	44	575	39	589	49	504	31	538	37
0	0.5	46	66	34	49	73	66	47	64	52	72	36	45	44	63

TABLE 3 (Continued)

		TEST													
$\delta\beta_0$	$\delta\beta_1$	A		B		C		D		D'		E		E'	
		$H_O(4)$	$H_O(3)$	$H_O(4)$	$H_O(3)$	$H_O(4)$	$H_O(3)$	$H_O(4)$	$H_O(3)$	$H_O(4)$	$H_O(3)$	$H_O(4)$	$H_O(3)$	$H_O(4)$	$H_O(3)$
0	1.0	52	103	34	84	111	103	49	102	51	113	34	77	43	96
0	1.5	48	183	36	146	184	183	42	176	48	185	34	139	40	161
0	2.0	43	293	30	238	274	293	43	268	42	276	28	228	33	242
0.5	0.5	97	66	81	55	251	66	92	67	95	75	68	52	83	60
1.0	1.0	179	93	143	71	660	93	170	86	185	88	135	63	162	74
1.5	1.5	386	197	320	149	780	197	364	182	369	192	307	145	333	167
2.0	2.0	607	275	537	227	723	275	584	268	601	269	524	222	556	252

8. Relation to the Literature

In Sec. 2.1, 2.2 known results are summarized. The results in sec. 2.3, 2.4 are thought to be new. The method of section 3 has been proposed in Walker and Lev (1953) but a different critical constant is used there, and the method is not investigated further there, nor is the method seen as a part of a family of tests, as in sec. 3.2, 3.3. The tests in §4 are known in the literature, although the user is not always warned of the effect of simultaneous tests on the overall significance level in older literature. In §5 known theorems are applied to the present case, apparently for the first time.

Related studies have been reported in the literature, e.g. by Erlander and Gustavsson (1965) where confidence regions for the difference between two regression lines are discussed. The critical constant found in that study is $4F(4, n_1 + n_2 - 4, \alpha)$ compared to $2F(2, n_1 + n_2 - 4, \alpha)$ in this case. Comparison of regression lines when two dependent and two independent variables have been measured on the same subjects are described by Finney and Sampford (1961). The reader is also referred to Williams (1959) chapters 7, 8 and 9.

REFERENCES

- BENNETT, B.M. (1963). On tests concerning lines and planes in hyperspace, *Metrika* 7, 41-46.
- BOX, G.E.P. and MULLER, M.E. (1958). A note on the generation of random normal deviates. *Ann. Math. Statist.* 29, 610-11.
- BROWNLIE, K.A. (1960). *Statistical theory and methodology in science and engineering*. John Wiley, New York.
- CHOW, G.C. (1960). Tests of equality between sets of coefficients in two linear regressions. *Econometrica* 28, 591-604.
- ERLANDER, S. and GUSTAVSSON, J. (1965). Simultaneous confidence regions in normal regression analysis with an application to road accidents. *Rev. Int. Statist. Inst.* 33, 364-377.
- FINNEY, D.J. and SAMPFORD, M.R. (1961). The comparison of linear regressions for sets of correlated variates. *Aust. J. Statist.* 3, 1-19.
- FRIEDMAN, M. (1937). Testing the significance among a group of regression equations. *Econometrica* V, 194-5.
- JANSSON, B. (1966). *Random number generators*. Victor Pettersons Bokindustri Aktiebolag, Stockholm.
- KASTENBAUM, M. (1959). A confidence interval on the abscissa of the point of intersection of two fitted linear regressions. *Biometrics* 15, 323-4.
- MILLER, R.G. (1966). *Simultaneous statistical inference*. McGraw-Hill Book Co., New York.

TOCHER, K. (1952). On the concurrence of a set of regression lines. *Biometrika* 39, 109-117.

WALKER, H.M. and LEV, J. (1953). *Statistical inference*. Henry Holt and Co., New York.

WELCH, B.L. (1947). The generalization of Student's problem when several populations are involved. *Biometrika* 34, 28-35.

WILLIAMS, E.J. (1959). *Regression analysis*. John Wiley, New York.

Manuscript received 22nd November 1967.